

Probability

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1 Introduction

A probability distribution $P : \mathcal{P}(\Omega) \rightarrow [0, 1]$ is a function such that:

$$(1) P(\Omega) = 1.$$

$$(2) \text{ If } A_1, A_2, \dots, A_n \text{ is a partition of } A, P(A) = P(A_1) + P(A_2) + \dots + P(A_n).$$

$$P(\emptyset) = 2P(\emptyset) \rightarrow P(\emptyset) = 0.$$

$$P(A \cup B) = P(A - B) + P(B - A) + P(A \cap B). \quad P(A) = P(A - B) + P(A \cap B) \\ \text{and } P(B) = P(B - A) + P(A \cap B), \text{ so } P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

$$\text{If } A \subset B, \text{ then } P(B) \geq P(A). \quad P(B) = P(A) + P(B - A) \geq P(A).$$

$$\sum_{i=1}^n P(A_i)P(B|A_i) = \sum_{i=1}^n P(A_i \cap B) = P(\Omega \cap B) = P(B).$$

Let $P(B) \neq 0$ and $Q : \mathcal{P}(B) \rightarrow [0, 1]$ be defined as $Q(A) = P(A|B)$. Then Q is a distribution on B .

$$Q(A) \in [0, 1]. \quad Q(B) = P(B|B) = P(B)/P(B) = 1.$$

$$Q(A) = P(A)/P(B) = \frac{1}{P(B)} \sum_i P(A_i) = \sum_i P(A_i|B) = \sum_i Q(A_i).$$

$$\begin{aligned} P(A \cup B \cup C) &= P(A \cup B) + P(C) - P((A \cup B) \cap C) \\ &= P(A) + P(B) + P(C) - P(A \cap B) - P((A \cap C) \cup (B \cap C)) \\ &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C). \end{aligned}$$

$P(A \cup B) \geq P(A) + P(B) - P(A \cap B)$. Assume

$$P(\bigcup_i A_i) \geq \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j).$$

$$\begin{aligned} P(\bigcup_i A_i \cup A_{n+1}) &= P(\bigcup_i A_i) + P(A_{n+1}) - P(\bigcup_i A_i \cap A_{n+1}) \\ &\geq \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + P(A_{n+1}) - \sum P(A_i \cap A_{n+1}) \\ &= \sum_{i=1}^{n+1} P(A_i) - \sum_{1 \leq i < j \leq n+1} P(A_i \cap A_j). \end{aligned}$$

$P(A|B) = P(A|B^C)$. If $P(B) = 0$, then $P(A \cap B) \leq P(B) = 0$, so $P(A \cap B) = 0$. If $P(B^C) = 0$, then $P(A \cap B^C) = 0$, so $P(A \cap B) = P(A)$. Also, $P(B) = 1$. Else,

$$\begin{aligned} P(A \cap B)/P(B) &= P(A \cap B^C)/P(B^C) \\ &= \frac{P(A) - P(A \cap B)}{1 - P(B)}, \end{aligned}$$

so $P(A \cap B) - P(A \cap B)P(B) = P(A)P(B) - P(A \cap B)P(B)$, so $P(A \cap B) = P(A)P(B)$.

Easier: $P(A) = P(A|B)P(B) + P(A|B^C)P(B^C) = P(A|B)$.

If A, B are independent, then so are A, B^C . $P(A \cap B^C) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)P(B^C)$.

If $\{A, B\}, \{A, C\}$ are independent, then so is $\{A, B \cap C\}$. False.